

SASH-VPV Palm Vein Biometric Platform — Business Plan

Product: Secure Authentication via Subcutaneous Vascular Palm-Veins (SASH-VPV)
Institution: National University of Technology (NUTECH), Islamabad
Market: Pakistan — metro transit, banking, fintech, BPO, corporate, co-working, and adjacent sectors
Document purpose: Strategic roadmap for commercialization, NADRA alignment, and sector-by-sector deployment

1. Executive Summary

SASH-VPV is a contactless palm vein recognition platform built on the XRTECH MagicVein Plus NIR scanner, a custom-trained neural matcher (EfficientNet-B0 + CBAM, 512-d embeddings), and a full-stack admin/member portal. The system already supports live enrollment, 1:1 verification, 1:N identification, audit logging, and role-based access.

Commercial thesis: Palm vein biometrics are more hygienic, harder to spoof, and more privacy-preserving than fingerprints — especially relevant in Pakistan where fingerprint systems face wear-and-tear issues (manual labor, aging skin, dry climate) and public hygiene concerns post-COVID.

Primary opportunities:

- 1. **Metro / mass transit ticketing** — mobile transit app with prepaid wallet; palm scan at any gate auto-deducts fare
- 2. **NADRA-aligned digital identity** — national ID verification layer for high-trust services
- 3. **Enterprise access & workforce** — banks, fintech, BPOs, corporate offices, co-working spaces (web admin portal)

Recommended positioning: *"Pakistan's contactless vein identity layer — more authentic than fingerprint, built for high-throughput public and financial environments."*

2. Problem Statement (Pakistan Market)

Pain point	Fingerprint limitation	Palm vein advantage
Worn/damaged prints (field workers, elderly)	High false-reject rate	Vein pattern is internal; skin surface condition matters less
Hygiene in public/shared devices	Contact required	Contactless (3–8 cm hover)
Spoofing (latent prints, molds)	Easier to lift/replay	Liveness via NIR + subdermal pattern
Religious/cultural touch aversion	Physical contact	No touch

High-throughput gates (metro, offices)	Slower, needs cleaning	Faster flow, no residue
Data sensitivity	Fingerprint widely leaked in legacy DBs	Vein templates harder to reverse-engineer; different regulatory narrative

Pakistan's digital identity stack (CNIC, SIM verification, Raast, branchless banking) is mature at the **document/number** level but weak at **continuous physical authentication** in daily high-volume settings. SASH-VPV fills the gap between *"prove once at onboarding"* and *"authenticate every day at speed."*

3. Product Foundation (What You Have Today)

Layer	Capability	Business relevance
Hardware	XRTECH MagicVein Plus, 480×640 NIR, USB	Deployable gate scanner / kiosk module
SDK + capture	Live MJPEG stream, distance sensing, auto-heal	Operator-friendly; reduces downtime
AI matcher	SASH-VPV embedder, cosine similarity, quality gates	Custom IP; not vendor-locked to SDK match only
Web portal	Admin, member, employee roles; enrollment; logs	Enterprise / BPO / corporate deployments
Transit mobile app	Passenger wallet, top-up, trip history, palm-linked account	Primary commuter interface for metro/BRT
Dataset	2,667 images, 122 subjects (SASH-VPV corpus)	Research credibility; retrain path for local demographics
Security	JWT auth, template storage, audit trails	Baseline for SOC/compliance conversations

Gap to production (honest): Scale testing on Pakistani demographics, PAD (presentation attack detection) certification, offline edge deployment, and formal NADRA API integration are roadmap items — not blockers for pilots.

4. Strategic Pillars

Pillar A — Transit (Metro Ticketing via Mobile App)

Vision: Palm-in, ride — no card, no physical ticket. Passengers use a **dedicated mobile transit app** (iOS/Android), not a web portal, for everything except the scan at the gate.

Passenger mobile app (core product for transit):

- **Account & identity:** Sign up with CNIC + mobile → enroll palm (at station kiosk or partner center) → palm template linked to transit account
- **Wallet / balance:** Load balance via JazzCash, Easypaisa, Raast, debit card, or bank transfer
- **Auto fare deduction:** When passenger scans palm at **any** metro/BRT gate, the system:
 1. Identifies the user (1:N match at gate)
 2. Calculates fare (origin–destination or flat rate, per transit authority rules)
 3. **Automatically deducts** the fare from in-app balance
 4. Opens the gate and logs the trip in the app (timestamp, station, fare, remaining balance)
- **Trip history & receipts:** Full ride log, low-balance alerts, instant top-up push notifications
- **No phone at gate:** Passenger does **not** need to open the app to travel — palm alone triggers payment; the app is for wallet management and history

Gate / backend flow:

- **Gate hardware:** Palm vein scanner + turnstile at each entry/exit; 1:N identify in <1 s (GPU) or <3 s (CPU pilot)
- **Fare engine:** Central backend computes fare by entry/exit pair (or zone-based); rejects travel if balance insufficient
- **Interoperability:** One palm account works across all gates on the network — travel anywhere on the line/city without separate tickets
- **Fallback:** QR one-time ticket or NFC card if match fails or balance is zero

Why mobile app instead of web for transit:

- Push notifications (low balance, promo fares)
- Native payment SDK integration (easier top-up)
- Offline ticket cache / last-known balance view
- Higher daily engagement vs browser bookmark
- App Store presence = consumer trust for a public-facing product

Metro engagement path:

1. Launch transit app beta + pilot 2 gates at one station (e.g., Orange Line Lahore, Islamabad Metro, or Peshawar BRT)
2. Measure: top-up conversion, auto-deduction success rate, throughput (passengers/hour), false accept/reject
3. Expand to full station → line-wide → city-wide; single wallet valid on entire network

Revenue: Per-transaction fee (% of fare), mobile wallet float interest (where permitted), annual station license, hardware lease, transit authority integration contract, in-app promotions.

Pillar B — NADRA Collaboration

Vision: Palm vein as a **secondary biometric factor** bound to CNIC identity — not a replacement for NADRA's core systems, but an accelerator for private-sector trust.

Why NADRA would care:

- Reduces synthetic identity fraud in banking/fintech onboarding
- Contactless alternative where fingerprint quality is poor
- Aligns with Pakistan's digital public infrastructure narrative

Collaboration models (pick one to start):

Model	Description	Your role	NADRA role
Verification API	Merchant sends CNIC + live palm → NADRA returns match/no-match	Build capture SDK + relay	Identity authority, legal framework
Delegated enrollment	NADRA-certified centers enroll vein templates linked to CNIC	Operate enrollment kiosks	Certify process, hold golden record
Private template, public key	Vein template stored by you; NADRA holds hash/reference only	Template vault + HSM	Audit + dispute resolution
Pilot MOU	University/industry pilot under NUTECH + NADRA innovation channel	Research + prototype	Sandbox approval, data governance

Practical first step: Propose a **sandbox pilot** — 500–1,000 volunteers, CNIC-verified enrollment, independent evaluation report (EER, FAR, FRR on local population). NADRA responds to evidence, not slides.

Compliance talking points:

- Templates are mathematical embeddings, not images
- No palm image stored post-enrollment (configurable)
- Audit logs for every match attempt
- Data residency in Pakistan

Pillar C — Financial Services (Banks & Fintech)

Vision: Palm-based step-up authentication for high-risk actions.

Use cases:

- Branch login / privileged teller actions
- Fintech app: transfer above threshold, device re-bind, password reset
- ATM cardless withdrawal (palm + PIN)
- Merchant POS verification for high-value transactions

Approach:

1. **SDK/API integration** — banks embed your capture widget; your cloud matches against enrolled template
2. **On-prem appliance** — box at bank DC for air-gapped matching (regulatory preference)
3. **Co-brand with 1Link / Raast** — position as biometric step-up in payment rails

Why banks switch from fingerprint:

- Lower false rejects → fewer branch visits
- Better fraud story for SBP inspections
- Premium "vein secure" tier for corporate banking clients

Revenue: Per-seat license, per-transaction auth fee, annual support, hardware bundle.

Pillar D — BPO & Corporate Offices

Vision: Shift attendance, secure floor access, and clean-desk policy compliance — one palm scan.

Use cases:

- Time & attendance (replace fingerprint machines)
- Data-center / server room access
- Hot-desk booking verification in co-working spaces
- Visitor management: pre-enroll palm + CNIC for recurring contractors

Approach:

- **Turnkey kiosk:** Scanner + your portal (employee panel already exists)
- **HRIS integration:** Export attendance to SAP, Oracle, local HR tools via API
- **Multi-tenant SaaS:** One platform, many companies (co-working operators love this)

Revenue: Per-employee/month, per-site license, setup fee.

Pillar E — Co-working & Shared Spaces

Vision: Members scan palm at door; billing and access auto-sync.

Approach:

- Integrate with access control (HID, ZKTeco, Hikvision) via relay/API
- Member app: enroll once, use any location in network
- Day-pass: CNIC verify + single-palm enroll at front desk

Revenue: White-label for operators (Regus-style, local brands), rev-share on membership.

5. Sector Expansion Map

Sector	Entry use case	Expansion	Sales cycle
Metro / BRT	Mobile app + gate pilot	City-wide palm wallet network	12–24 months
Banks	Branch pilot	API + ATM	9–18 months
Fintech	Step-up auth	Platform SDK	3–6 months
BPO	Attendance	Full access control	3–6 months
Corporate	Single HQ	Multi-site	6–12 months
Co-working	One location	Chain rollout	1–3 months
Healthcare	Staff access	Patient check-in	12+ months
Education	Campus attendance	Exam hall identity	6–12 months

Government offices	Citizen service queue	e-gov kiosk	18+ months
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Fastest revenue: BPO, co-working, fintech (short cycles, less procurement).

Highest impact: Metro + NADRA (credibility + scale).

Highest margin: Bank on-prem appliance + support.

6. Go-to-Market Strategy

Phase 1 — Proof (Months 1–3)

- Fix production deployment (GPU inference, gate hardware enclosure)
- **Transit mobile app MVP:** wallet, top-up, trip history, palm enrollment flow
- Run public demo at NUTECH + 1 corporate partner
- Publish Pakistan-specific accuracy benchmark (new local captures)
- File provisional patent / copyright on SASH-VPV pipeline

Phase 2 — Pilots (Months 4–9)

- **Transit:** mobile app live + 1 metro station gate pilot (auto fare deduction)
- OR 1 bank branch OR 3 BPO sites (pick two, not all three)
- NADRA sandbox MOU application
- Hire 1 enterprise sales + 1 integration engineer + 1 mobile developer

Phase 3 — Scale (Months 10–18)

- SaaS multi-tenant portal
- Certified enrollment partners
- Channel partners (access control vendors, POS integrators)

Messaging by audience

Audience	Lead message
Transit authority	Mobile wallet + auto fare deduction; throughput + fare evasion reduction
Commuter (B2C)	Load once, travel anywhere — just show your palm
NADRA	Fraud reduction + contactless national identity layer
Bank CISO	Spoof resistance + audit trail + SBP alignment

Fintech founder	Drop-in SDK, faster than building in-house
BPO HR	Zero-touch attendance, no fingerprint maintenance
Co-working ops	Premium member experience, one scan

7. Revenue Models

Model	Example pricing (PKR, indicative)	Best for
Mobile wallet float	Revenue share on idle balance	Transit app
Per-transaction (% of fare)	1–3% per ride	Metro, BRT
SaaS per user/month	150–500 / user / month	BPO, corporate, co-working
Site license (annual)	500K–5M / site	Bank branch, metro station
Hardware + software bundle	80K–200K per kiosk	Turnkey gates
Integration project	1M–10M one-time	Enterprise, government
NADRA/certification services	Custom	National programs

Target Year 1 (conservative): 3 paid pilots + 2 recurring BPO/co-working contracts → fund Phase 2.

8. Technical Deployment Strategies

Strategy 1 — Cloud API (fastest)

- Your FastAPI backend on Azure Pakistan / local DC
- Clients call REST: enroll, verify, identify
- Pros: fast integration; Cons: latency, data residency concerns

Strategy 2 — On-prem edge box

- Mini PC + scanner at each site; sync gallery nightly
- Pros: bank/gov friendly; Cons: ops overhead

Strategy 3 — Hybrid (recommended for metro)

- **Mobile app + cloud:** Wallet, top-up, and account management in transit app; palm templates and balance in central DC
- **Edge at gate:** 1:N match locally for speed; fare deduct API call to central wallet service
- Best for metro: central account, local gate speed, auto deduction on every scan

Strategy 4 — Mobile transit app + fixed enrollment kiosk

- **Enrollment:** Fixed station kiosk (quality-controlled palm capture) or partner center
- **Travel:** Palm scan only at gate — no phone needed; fare auto-deducted from app wallet
- **Management:** Passenger uses mobile app for balance, history, and top-up only

Hardware BOM (per gate/kiosk, indicative):

- XRTECH scanner: vendor quote
- Industrial mini PC (GPU optional): 80K–250K PKR
- Enclosure + mount: 30K–80K PKR
- Networking + UPS: 20K–50K PKR

9. NADRA & Regulatory Roadmap

1. **Legal review** — Personal Data Protection Bill alignment; consent flows for biometric collection
2. **Biometric policy** — Document template format, retention, right to erasure
3. **NADRA engagement** — Innovation cell / MOU for pilot; never claim "NADRA approved" without written consent
4. **SBP (banks)** — Biometric authentication guidelines for digital banking
5. **PTA** — If SIM re-verification angle is pursued, separate track
6. **Certification path** — ISO/IEC 30107 (PAD), ISO 19794 (biometric data interchange) as long-term goals

10. Competitive Positioning vs Fingerprint

Sell these five facts:

1. **Internal pattern** — veins are subdermal; harder to copy than surface prints
2. **Contactless** — faster lines, better hygiene, culturally easier
3. **Lower maintenance** — no optical platen cleaning; fewer "finger not recognized" support tickets
4. **Dual-hand enrollment** — your system already supports Left/Right templates per identity
5. **Custom AI** — SASH-VPV model is yours; not dependent on vendor black-box matching alone

Objection handling:

- *"Fingerprint is cheaper"* → TCO: support calls + cleaning + false rejects often exceed hardware savings
- *"We already have NADRA fingerprint"* → Position as **complement** for daily auth, not CNIC replacement
- *"Is it proven?"* → Cite NUTECH dataset, pilot metrics, international vein literature (Fujitsu, Hitachi)

deployments in Asia)

11. Partnership Strategy

Partner type	Target examples	Value exchange
Scanner OEM	XRTECH / local distributor	Volume pricing, co-marketing
Access control	ZKTeco, Hikvision integrators	Your AI + their locks
Payments	1Link, JazzCash, Easypaisa	Transit app top-up + wallet rails
System integrators	TRG, NetSol, local SIs	They sell, you deliver API
Universities	NUTECH, NUST, COMSATS	Talent + research credibility
Government	NADRA, transit authorities	Pilots + policy alignment

12. 18-Month Roadmap

Quarter	Milestone
Q1	Production hardening; transit mobile app MVP; 2 corporate pilots; NADRA sandbox proposal
Q2	Metro station pilot (palm + auto fare deduction); publish local accuracy whitepaper
Q3	Multi-tenant SaaS; 5 paying sites; channel partner signed
Q4	NADRA pilot results; scale first vertical; seek seed/grant funding
Q5–Q6	Second city / second bank; on-prem appliance SKU; team of 8–12

13. Team & Funding

Core team needed:

- CEO / business development (gov + bank + transit relationships)
- CTO (platform + security)
- **Mobile developer** (transit app — iOS/Android)
- ML engineer (retrain, PAD)

- Integration engineer (payment APIs, transit fare engine, access control)
- Field ops (gate install, support)

Funding sources:

- HEC / Ignite / NCAI grants
- NUTECH incubation
- Angel (fintech founders)
- Strategic (access control vendor)
- Revenue from pilots (preferred — proves market)

14. Risks & Mitigations

Risk	Mitigation
NADRA timeline slow	Run commercial pilots without NADRA badge; add later
Hardware supply	Lock distributor agreement; keep 30-day spare stock
False match in wild	Quality gates (already built); human fallback; threshold tuning per sector
Privacy backlash	Consent-first enrollment; no image retention option; local hosting
Incumbent fingerprint vendors	Integrate alongside; sell upgrade path
GPU cost at gates	Edge TPU / model distillation; 1:N cache per station

15. Immediate Action List (Next 30 Days)

1. Package a **10-slide pilot deck** (metro mobile app + bank versions)
2. Record a **2-minute demo video** — app top-up → palm scan at gate → auto fare deduction → trip in app
3. Write **NADRA sandbox proposal** (1 page problem, 1 page method, 1 page ask)
4. Contact **1 BPO + 1 co-working space** in Islamabad/Rawalpindi for free 30-day pilot
5. Add **GPU inference** to production backend for gate-speed matching
6. Register **business entity** (SMC / Pvt Ltd) and open business bank account
7. Define **pilot SLA**: uptime, match latency, support response

16. Success Metrics (KPIs)

KPI	Pilot target	Production target
Wallet top-up success rate	> 95%	> 99%
Auto fare deduction success	> 98%	> 99.5%
False Accept Rate (FAR)	< 0.1%	< 0.01%
False Reject Rate (FRR)	< 5%	< 2%
Match latency	< 3 s	< 1 s
Enrollment time	< 90 s	< 60 s
Gate throughput	20/min	40+/min
Uptime	99%	99.9%
User satisfaction (NPS)	> 40	> 60

17. One-Line Pitch

"SASH-VPV is Pakistan's contactless palm vein platform — load balance in the transit app, scan your palm anywhere, and fare deducts automatically. More reliable than fingerprint, with a clear path to NADRA-aligned national trust."

Document version: 1.1 — transit channel is mobile app (wallet + auto fare deduction); web portal retained for enterprise/admin. Aligned with SASH-VPV codebase. Update quarterly as pilots produce real metrics.